

MADYANN SAIDI

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EDUCATION

UNIVERSITY OF MARYLAND – College Park

May 2028

4.00 GPA - B. S. Mechanical Engineering

QUEST Honors Program

- A selective, multidisciplinary honors program focused on systems thinking, teamwork, and project management.

RESEARCH & INTERN EXPERIENCE

UMD's XPRIZE WILDFIRE COMPETITION TEAM

College Park, MD

Drone Software Intern

Aug 2025 – Present

- Develop an incipient fire detection model using 1,230+ manually annotated images and 4 datasets, achieving over 95% smoke/fire detection accuracy.

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

Gaithersburg, MD

Research Fellow

May – Aug 2025

- Created an ADMET test setup and analyzed human-robot contact pressure distribution on 7 cylindrical objects, providing critical cross-validation data to refine FEA simulation precision.
- Designed a MATLAB-based closed-loop analysis tool for Pressure Sensitive Film results, reducing data analysis time ~30× and offering ~100+ customizable parameters for thorough testing/analysis.
- Designed 3 bimanual manipulation test procedures on an ABB Yumi robot using RobotStudio, enabling 6-axis force/torque data capture. Programmed ABB Yumi to autonomously solve a 2x2x2 Rubik's Cube in ~20-30 min.

Research Assistant/Intern

Jun 2023 – Jun 2024

- Redesign ICRA manipulation board in SolidWorks into 4 modular quadrants to optimize layout and manufacturability, increased task complexity 1.5×, and reworked a 12-page manual utilized by university teams.
- Optimized the Yale OpenHand Model T in SolidWorks to improve its durability by 5× and boost grasp strength from 11 N to 34 N (~3×), receiving commitment for redesign inclusion in future iterations.

PROJECTS & LEADERSHIP ROLES

Mechanical Officer - QUADRUPEL ROBOTICS CLUB (Testudog)

Jun 2025 – Present

- Lead the mechanical subteam of a 30+ member robotics organization by coordinating design, prototyping, and testing efforts to advance quadruped robot balance, dynamics, and hardware software integration.

Project Manager/Lead Programmer - SEED PLANTING OTV (Over-Terrain Vehicle)

Jan – Jun 2025

- Directed 8-person team through Trello and weekly meetings to develop autonomous Arduino-based OTV; achieved 100% mission success, collecting 200% minimum payload in under 3 minutes; awarded Best Mission Award.
- Engineered a full autonomy pipeline using ArUco marker localization, PID control tuning, and a calibrated color sensor to achieve 95% classification accuracy for seed planting and rock retrieval missions.

Designer - PROSTHETIC HAND, PROJECT M²

Sep 2024 – Jan 2025

- Accomplished 32 unique finger configurations and ~90% functional dexterity by designing and iterating a pulley-driven single-motor prosthetic hand in SolidWorks.

CAD Design Lead - FRC ROBOTICS - TEAM 1389

Sep 2023 – Jun 2024

- Mentored 15+ teammates in CAD modeling and design-for-manufacturing principles; improved team design collaboration efficiency by 50% through structured training and peer feedback sessions.
- Led mechanical chassis redesign and integrated a 4-roller shooter with an autonomous ArUco vision aiming system; increased accuracy by 85% and durability 150%, reducing on-site repairs during competitions.

SKILLS

Skills: SolidWorks (CSWA & CSWP), MATLAB, Python, Java, ABB RobotStudio, Arduino, robotics programming, vision systems, machine learning, leadership, team collaboration, project management.

Languages: English (native), French (fluent), Lebanese Arabic (proficient)

PUBLICATIONS

Finite Element and Empirical Assessment of Pain-Related Stress in Human-Robot Contact Based on Tool Geometry
– *International Journal of Robotics & Automation (Manuscript Submitted)*

2025